

↑ Agent PDF

- ① Human Agent
- ② Robotic Agent
- ③ Software Agent

Perception

PEAS → ~~Re~~ Performance
Environment
Actuators
Sensors

↑ Specifying the Task Environment

{ Description from PEAS }

Agent Type	Performance Measure	Environment	Actuators	Sensors
Taxi Driver	Safe, Fast, Legal, Comfy, Max profit	Roads, Other traffic, Pedestrians, Customer	Steering, Acceleration, Brake, Signal, Horn, Display	Cameras, Sonar Speedometer GPS, Odometer, Accelerometer, Engine Sensors, Keyboard

Agent Type	Performance Measure	Environment	Actuators	Sensors
Medical Diagnosis System	Healthy patient, Reduced costs	Patient, Hospital, Staff	Display questions, tests, diagnoses, treatments, referrals	Keyboard entry of symptoms, findings, patient's answer

Internet Shopping Agent

Performance Measure : Price, Quality, Appropriateness, Efficiency.

Environment : Current & Future WWW sites, vendors, Shippers.

Actuators : Display to user, follow URL, fill in form

Sensors :

Knowledge

- ▷ Can be defined as the body of facts & principles accumulated by humankind or the act, fact, or state of knowing
- ▷ It is having a familiarity with language, concepts, procedures, rules, ideas, abstractions, places, customs, facts & associations, coupled with an ability to use these notions effectively in modeling different aspects of the world.

→ Types of Knowledge

① Declarative:

Passive Knowledge expressed as statements of facts. "Theoretical"

② Procedural: Compiled knowledge related to the performance of some tasks. "Practical"

③ Heuristic:

"Special type of Knowledge"

The knowledge used to make good judgements or strategies, tricks, or 'rule of thumb' to simplify the solution of problem.

⇒ may not always be correct.

→ Knowledge & Data

Knowledge $\neq$ Data

→ Belief: Meaningful & coherent expression [Maybe true/false]

→ Hypothesis: A justified belief that is not known to be true (may be false)

→ Knowledge: True justified belief.

↑ Environment Types

① Fully Observable

② Partially Observable

↳ noisy & inaccurate sensors

↳ missing states from sensor data

③ Deterministic → determined by the current state & action

④ Stochastic → outcome of uncertain actions & probability

⑤ Strategic → establish clear goals & objectives

⑥ Episodic : each episode depends only on the episode

⑦ Sequential : the current decision could affect all future decision.

⑧ Static : Unchanged

⑨ Dynamic : Ever changing / ~~स्थिर~~

⑩ Discrete : Limited moves

⑪ Continuous : taxi drivers

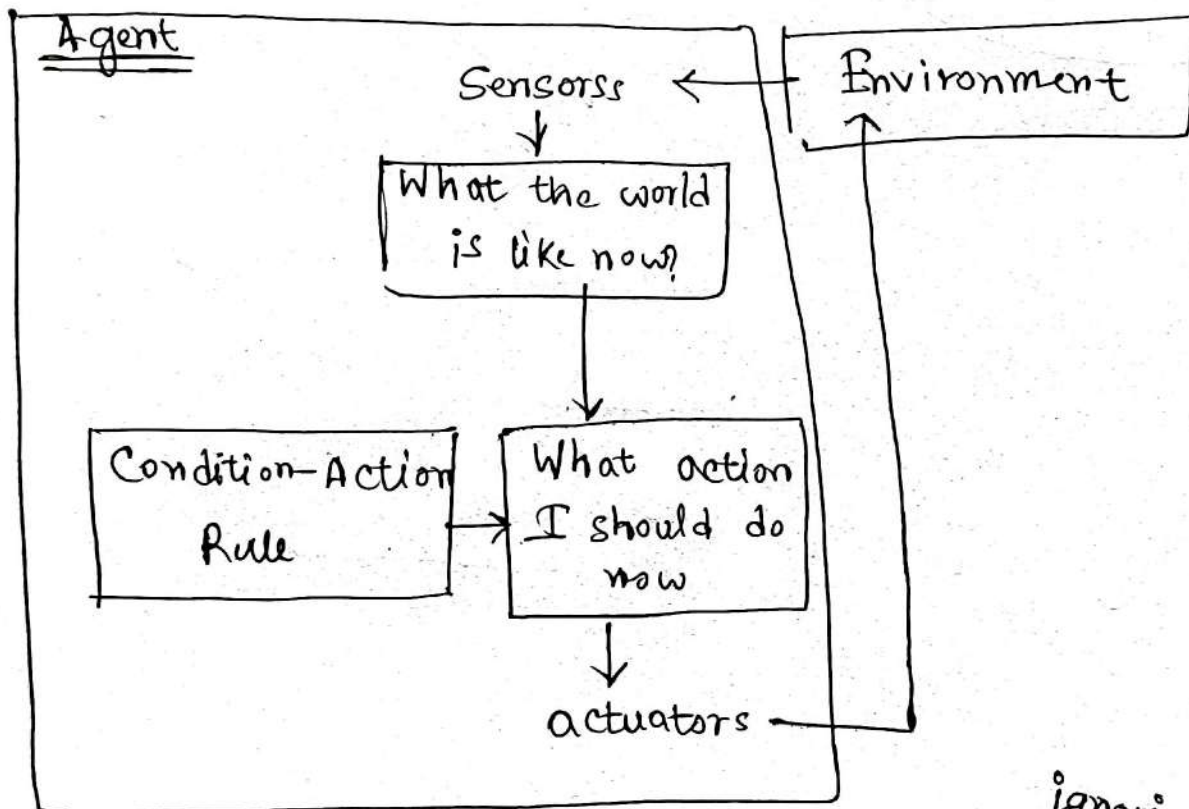
⑫ Single Agent :

⑬ Multiagent :

Types of Agent

Expect questions from this topic

- ① Simple reflex agents
- ② Model-based reflex agents
- ③ Goal-based agents
- ④ Utility-based agents



Basic function: based on current perception ^{ignoring} ~~gathering~~ historical percepts.

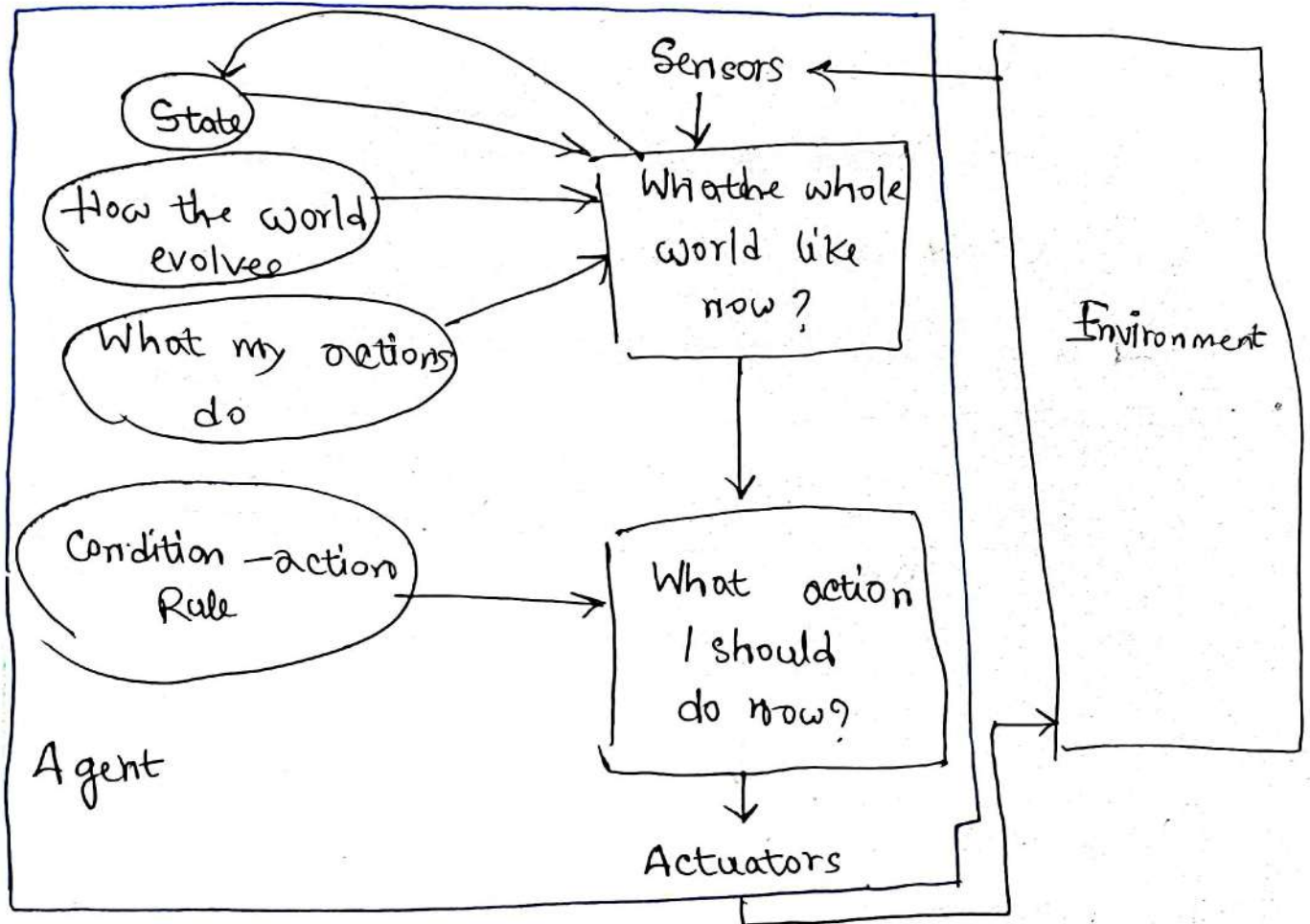
Condition-action rule: if-then rule.

Simplicity & Limitations: intelligence & memory is limited.

Problems with observability.

Simple Reflex Agent Example PDF

Model Based Reflex Agents



Goal-Based Agent

Goal oriented

Utility-Based Agent

Next class : Figure